DO NOW — RECALL

# Thinking logically: decisions and program flow

Name: _____ Class: _____ Date: _____

**1. Why must a booking be saved before confirmation?**

**2. What is true AND false?**

**3. What does an if condition control?**

**4. What is false OR true?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Confirmation must reflect a successful stored change.

### 2. Model answer

False.

### 3. Model answer

Which branch of the program runs.

### 4. Model answer

True.

### Vocabulary — know these terms

condition — Expression evaluated as true or false.

branch — Path selected by a decision.

AND — True only if both operands are true.

OR — True if either or both operands are true.

truth table — All Boolean input combinations and outputs.

NEW LEARNING

# Read, annotate and explain

### Locate the decision

A discount needs BOTH an eligible membership and a basket of at least £50.

### Follow the true branch

For member=true and basket=60, the condition is true and 10% discount produces £54.

### Follow the false branch

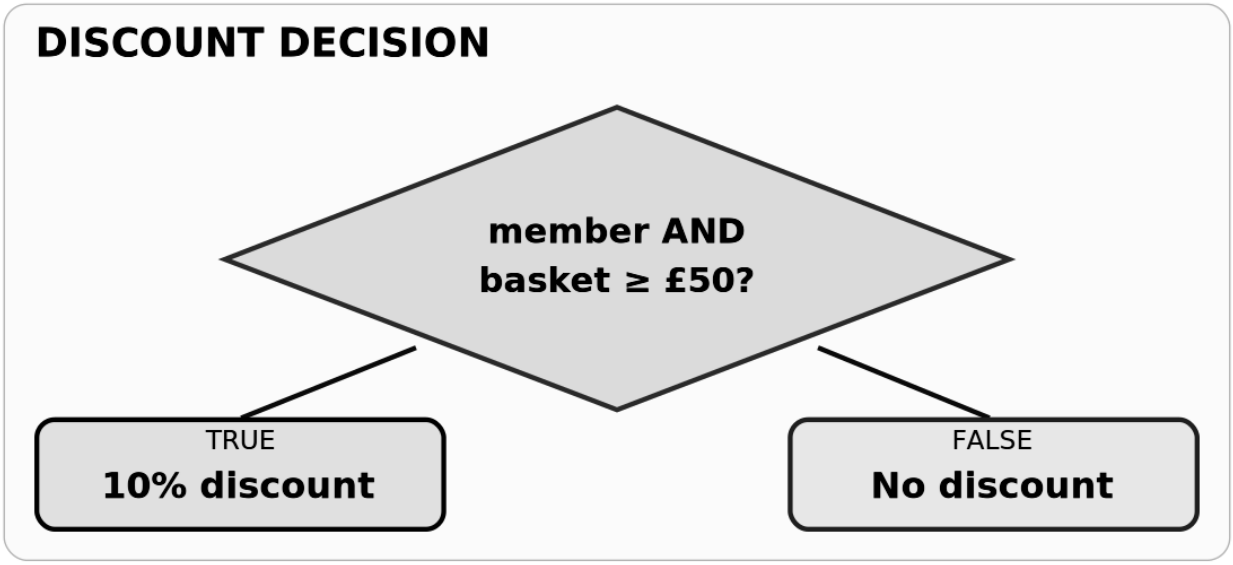
For member=false with basket=60, the AND expression is false, so the full £60 is charged.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Discount decision: member and basket at least fifty pounds. True gives ten percent discount; false gives no discount.

### New Learning




TRUE BRANCH; Member = true; Basket = £60; Charge = £54

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




FALSE BRANCH; Member = false; Basket = £60; Charge = £60

I DO — WORKED EXAMPLE

# Follow the worked method

**A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.**

### Predict first

The age condition is true, but the ticket condition is false.

### Model the reasoning

AND requires both true; the result is false, so the deny branch executes.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.

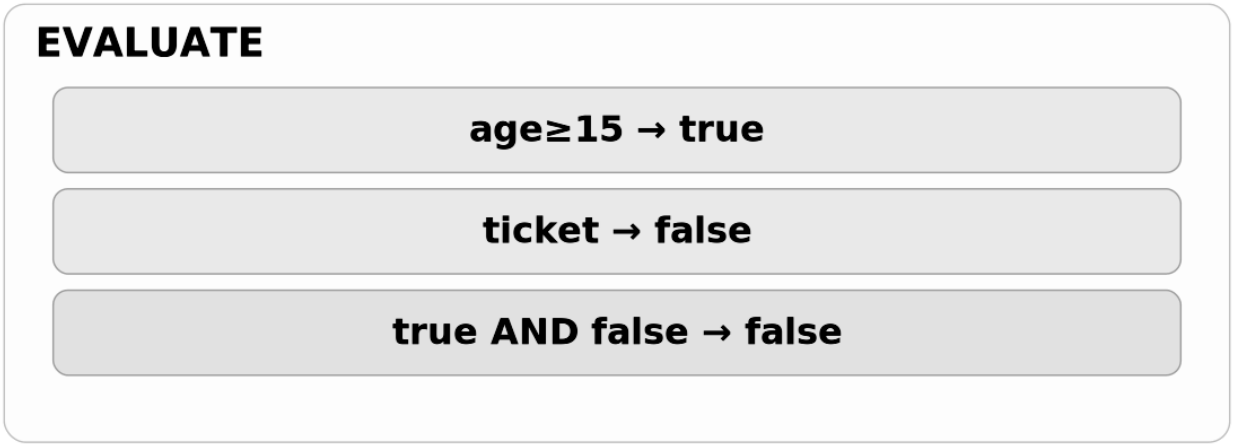
The age condition is true, but the ticket condition is false.




INPUTS; age = 16; hasTicket = false

### A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.

The age condition is true, but the ticket condition is false.



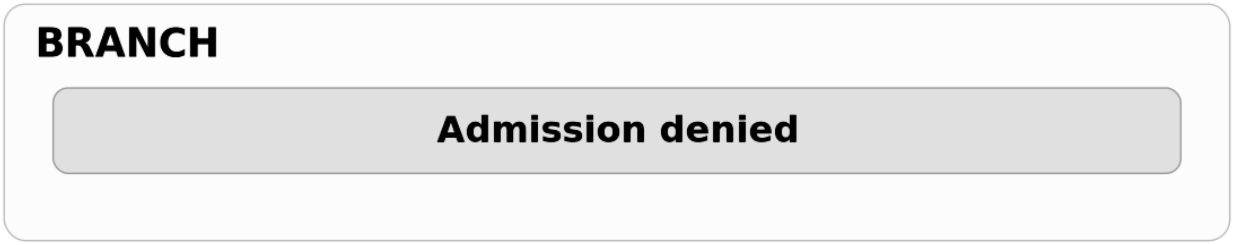

EVALUATE; age≥15 → true; ticket → false; true AND false → false

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A cinema permits admission for age≥15 AND a valid ticket. Trace age=16 with no ticket.

The age condition is true, but the ticket condition is false.




BRANCH; Admission denied

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A challenge unlocks when score≥80 OR teacherOverride=true. Trace score=72 with override.**

The score condition is false because 72<80.




INPUTS; score = 72; teacherOverride = true

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first The score condition is false because 72<80.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Give A AND B for (false,false), (false,true), (true,false), (true,true).

### Question 2

Access is allowed if Year13 AND (HasCode OR TeacherApproved). Evaluate Year13=true, HasCode=false, TeacherApproved=true; then Year13=false, HasCode=true, TeacherApproved=true.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

A developer changes the discount condition `member && basket>=50` to `member || basket>=50`. Give two wrong outcomes and correct the expression.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Replacing AND with OR in a two-part eligibility rule can permit extra cases.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

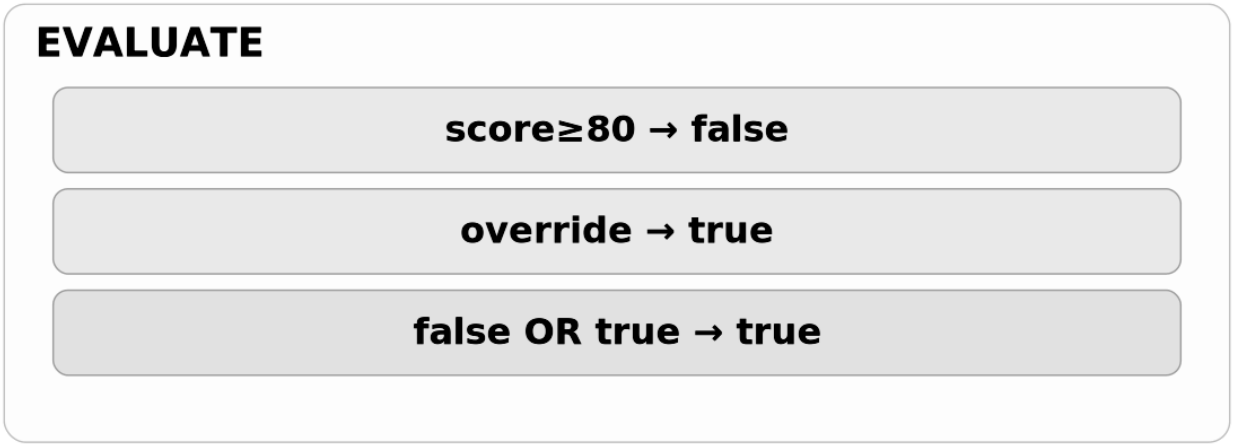
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A challenge unlocks when score≥80 OR teacherOverride=true. Trace score=72 with override.**

The score condition is false because 72<80.




EVALUATE; score≥80 → false; override → true; false OR true → true

**A challenge unlocks when score≥80 OR teacherOverride=true. Trace score=72 with override.**

The score condition is false because 72<80.




BRANCH; Challenge unlocked

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Model the reasoning

The override is true, so OR evaluates true and the challenge unlocks.

IMPROVE: Explain one correction to your own reasoning.